Competition Scenario (sunspire)

On a remote island far from most of the world lies the solarpunk city of Sunspire. A radiant forest-city that is composed of ecologically sustainable towns that blend technology and nature. Every home, garden, and communal hall is powered by a masterfully engineered computing network that provides the essential services necessary for society to function.

For the first time in its history, Sunspire would like to invest heavily in cybersecurity and kickstart its cyber servants program. A group responsible for ensuring the computer systems essential to Sunspire could never be destroyed. To fairly choose these newest civil servants, Sunspire is holding a tournament of strengths.

Sunspire has invited world-class cybersecurity teams from around the world to compete against each other for the chance to become the very first members of the Sunspire cyber servants program. Each team will compete against the others to break into Sunspire's network, stabilize it, and defend it from their competitors.

Kali Linux Machines

To connect to the competition environment and compete, use the existing Kali Linux machines. Each team's Kali IPs and username/password credentials are provided in their Discord channel, along with connection instructions. Some rules regarding the Kali machines are:

- **Do not** attempt to RDP into the Kali Linux machines outside of the normal competition hours. (during meetings and office hours)
- **Do not** delete or remove the following files on your Kali Linux machines:
    - `/home/kali/.bash_history`
    - `/home/kali/.full_command_history.csv`
    - `/root/.bash_history`
    - `/root/.full_command_history.csv`
- **Do not** change the default password or authentication information on your Kali machine. Specifically, the "kali" user.
- You may make minor changes to your Kali Linux machines yourself. Examples include:
    - Installing a new tool, programming language, or service
    - Changing the wallpaper
    - Adding new users/groups

You may request major changes or updates to your Kali Linux machines from the organizers. Examples may be:

- Update to a new version of the OS
- Expand disk space
- Changing the IP address

To make requests, please ping the organizers in your team's private Discord channel.

Competing

This section provides instructions for competing in the netsiege competition.

Overview and Scope

To compete, each team's goal is to gain control over as many of the competition servers as possible. Teams may not attack any servers that are not in scope. **As of phase 3 of the competition**, the only in-scope competition servers are at the following IP addresses:

- 10.100.7.40
- 10.100.7.41
- 10.100.7.42
- 10.100.7.43
- 10.100.7.44
- 10.100.7.45
- 10.100.7.46
- 10.100.7.47
- 10.100.7.48

There will be more in-scope servers as the competition and phases progress.

To gain control of a competition server, or a "box", teams must plant their token in any and all ctrl.txt files they can find. Tokens for your team can be collected from the scorekeeper service.

Every competition server has at least three users:

- Administrator/root: The default highest-privilege user on every machine
- "hostname": Every host has a user with the same name as the hostname of the machine. For example if the name of the host is "starfall" there is a user with a username of "starfall" on that machine.
- ops: This is a user account for the operations team (organizers). Please do not modify it.

Scoring

Scoring works in the following ways:

- Teams gain points by gaining and maintaining control over a box
- Control is gained by leaving your team's token in a ctrl.txt file on the box
- Tokens can be collected from the Scorekeeper engine once you log in as your team
- Teams gain points every X minutes for every server they control
- Teams will not gain points for control if the associated services with those servers are not accessible to the scoring engine.

- There may be multiple ctrl.txt files for each box; the more that are controlled, the more points you can gain. However, if you already control the “owner” ctrl file, meaning the highest privilege level possible (e.g. root/Administrator) then you will only gain points for that one.
- Multiple teams may not control different privilege levels on a machine (e.g., user vs root).

Scorekeeper and Tokens

Scorekeeper is each team’s one-stop shop for all things scoring-related. Scorekeeper is only accessible via the ThreatNet VPN (TCP). The scorekeeper scoring engine will be available at the following address:

- http://10.100.7.8/dashboard

Teams may log in to the scoring engine on either their Kali Machine or their personal computers. However, we suggest logging in via the Kali Machine so you can easily copy and paste your token from the scorekeeper page into your terminal (or whatever medium is appropriate).

Login information is available in your team’s Discord channel.

Tokens are unique UUID strings that will rotate every X seconds. You do not need to replace the contents of ctrl.txt every time it rotates. You must only replace it if another team places their data in and overwrites yours. The data in the ctrl.txt will naturally also be overwritten with nothing by the “tally” service. Don’t worry, this is normal behavior and just ensures that your token does not get stolen or abused by the next team.